

Clovis SFEIR

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Profile

IMT Atlantique engineering student specialising in quantitative modeling, statistical signal processing, and applied ML for industrial/financial systems. Built five systems (Python/Rust/C++20) benchmarked against industry references or labeled ground-truth: a from-scratch BM25 retrieval engine (**MRR 0.944**), an IIoT predictive-maintenance pipeline (**0/10 false positives**), an Ornstein-Uhlenbeck crude-spread engine, and a finite-difference PDE solver verified to **2nd-order convergence** against a closed-form solution.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*
Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France *Class rank: 1/85 | Software Engineering, Databases, Cybersecurity*
HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placement force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePLace (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

Engineering Docs RAG

Python, BM25, information retrieval

- From-scratch Okapi BM25 retrieval engine (chunking + ranking) for engineering documents (datasheets, HAZOP-style procedures); pluggable extractive/LLM generator interface, zero paid API calls in tests/CI/demo by design.
- Hand-derived BM25 scoring verified to 9 significant figures; measured **precision@1 0.90**, recall@5 1.00, **MRR 0.944** on a 40-question labeled set; ~3,330 docs/s ingest; 53 tests, zero runtime dependencies.

IIoT Predictive Maintenance

Rust, tokio, EWMA/CUSUM

- Async telemetry pipeline (tokio mpsc) fuses EWMA + CUSUM control charts across 3 sensor channels to detect developing bearing-wear signatures, with linear-trend remaining-useful-life estimation.
- Sensor-fusion tuning (2-of-3 channels + widened control limits) cut false positives from a 0.41–0.50 precision baseline to **0/10 across 10 seeds**; precision **1.0000**, recall 0.9642, F1 0.9818; **1.6 M samples/s**; 26 tests, 0 failures.

Crude Oil Spread Trading Engine

C++20, stochastic processes, order book

- Models refining margins (3:2:1 crack spread, generalized N:M:K) and mean-reverting inter-crude spreads via an Ornstein-Uhlenbeck process (Euler-Maruyama) with a rolling z-score signal, matched by a price-time-priority order book.
- OU stationary distribution verified against theory (500K-step sim); crack-spread formula checked to 10^{-9} ; order-book submit at **720 ns** (non-crossing)/**1.24 μ s** (crossing); 58 assertions/21 tests, ASan+UBSan clean.

Reservoir Diffusivity PDE Solver

C++20, OpenMP, finite-difference

- Explicit FTCS and implicit Crank-Nicolson finite-difference schemes – the same discretization family used for parabolic pricing PDEs – solving the reservoir diffusivity equation, grid-refinement convergence study against the closed-form erfc analytical solution.
- Error ratio converges to **4.0** as dx halves (verified 2nd-order accuracy, not assumed) across 5 refinements; measured (not assumed) OpenMP scaling identifies the Amdahl's-law-bounded sequential solve; 13 test cases, 40,300 assertions – the validate-against-ground-truth discipline any model needs before it's trusted.

HFT Limit Order Book Matching Engine

C++20, cache-aware layout, intrusive data structures

- Price-time-priority matching engine (limit/market/cancel) with intrusive doubly-linked price-level lists (**O(1) cancel** via pointer unlink) and pre-allocated object pooling to keep the hot loop free of `malloc/new`.
- Benchmark suite across three synthetic order-flow regimes: **2.3M+ msg/s** single-threaded throughput, **P50 0.6–1.0 μ s** / P99 2–4 μ s latency, **zero hot-path allocations**; unit tested.

Work Experience

Infotel

AI Engineer Apprentice – Lab IA

Nantes, France
Sep 2025 – Jun 2028

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests; validated end-to-end producing a fully working, 100%-test-passing application directly from a single specification.

Excelia Group

Software Developer Intern / AI Software Engineer Intern

La Rochelle, France
Two internships: Jan–Mar 2025 & May–Jul 2024

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + async parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

ML / Quant

BM25/information retrieval, EWMA/CUSUM change-detection, Kalman filter, Ornstein-Uhlenbeck, finite-difference PDE, LightGBM/XGB/CatBoost, FAISS

Languages

Python, C++20, Rust, TypeScript, SQL

Perf / Systems

OpenMP, cache-aware layout, intrusive linked lists, acquire/release atomics

Infrastructure

Docker, tokio async, Prometheus/Grafana, GitHub Actions CI, GCP, Redis, ZeroMQ

Frameworks

Flask, Node.js/Express, React/Next.js, REST/gRPC